# What Beliefs are Central in the Climate Change Belief System Network?

## Introduction

Climate change remains one of the most politically polarized issues, particularly in the United States. Sharp divides exist among the public and policy elites regarding the existence of climate change, the extent of its risk, and what, if anything, should be done to address it. A consequence of political divides regarding climate change is stymied progress in reducing greenhouse gas emissions.

There are several explanations for why the public remains divided on climate change, including political ideology and partisanship; cultural worldviews derived from cultural theory; environmental orientation; acceptance/rejection of the scientific consensus on climate change; and an aversion to government regulations meant to address climate change. Each explanation involves beliefs about climate change, yet these beliefs do not exist in isolation from one another. Beliefs about climate change form an interconnected belief system.

A belief system is a set of interrelated and connected beliefs. One way to understand belief systems is to view them as existing within a hierarchy, in which higher-order beliefs influence or constrain lower-order beliefs (Nohrstedt et al. 2023; Peffley and Hurwitz 1985; Sabatier and Jenkins-Smith 1993). With the hierarchical approach to belief systems, the most abstract beliefs and general principles, termed *deep core beliefs* (e.g., political ideology), inform less abstract views about policy issues, *policy core beliefs* (e.g., climate change is happening), as well as attitudes about specific policy approaches, or *secondary beliefs* (e.g., support for a carbon tax) (Sabatier and Jenkins-Smith 1993). A second way of understanding belief systems is that of a network of connected beliefs (Boutyline and Vaisey 2017; Brandt, Sibley, and Osborne 2019; Brandt and Sleegers 2021; Fishman and Davis 2022). Similar to other types of networks, belief system networks have *nodes*, which are the beliefs, and *edges*, which are the connections, or relationships, between beliefs. Belief system networks can be analyzed using centrality measures to understand the importance of the various beliefs (i.e., nodes).

Examining the public's climate change beliefs as a network can yield insights into polarized views. Indeed, several recent studies have examined the climate change belief system network (Lee et al. 2024, 2025; Lind, Morton, and Dalege 2024; Stewart, Rivera-Kientz, and Dacey 2024; Verschoor et al. 2020). However, apart from political ideology and partisanship, these studies did not include measures of other deep core beliefs such as cultural worldviews and environmental orientation, which have been shown to influence climate change beliefs (Jones 2011; Nowlin and Rabovsky 2020; Ziegler 2017).

In this paper, I use original survey data to examine the climate change belief system using measures of various types of beliefs not measured in previous research, including core beliefs (cultural worldviews, political ideology, partisanship, environment orientation), policy domain beliefs (views about climate change), and views on various policies to address climate change. By examining the climate change belief network, I can determine which beliefs are the most central within it. However, rather than testing competing hypotheses about which type of belief – deep core, policy core, or secondary – is most central, this paper takes an exploratory approach to mapping the structure of the climate change belief network. Prior research offers some expectations about which beliefs may take on structural importance. For example, hierarchical accounts of belief system structure point to deep core beliefs such as cultural worldviews and political identity. The "gateway belief" model points instead to perceived scientific consensus (e.g., Van Der Linden 2021), and solution-aversion accounts point to secondary beliefs and policy preferences. I use these three accounts as theoretical frameworks for interpreting which beliefs occupy structurally important positions in the network, rather than as mutually exclusive predictions to be tested. This allows the analysis to show how different kinds of beliefs are organized relative to one another.

I examine the climate change belief system network first as an undirected network, using a Gaussian graphical model (GGM), then as a directed Bayesian network. This allows for comparisons of centrality across an undirected and a directed network, the latter offering an estimated ordering among beliefs rather than a direct test of the theoretical deep core/policy core/secondary hierarchy. Using common measures of centrality – strength, closeness, and betweenness – I find that egalitarianism (a deep core belief), support for EPA regulations (a secondary belief), and climate change risk (a policy core belief) have the highest point estimates for centrality in the GGM, though pairwise comparisons show these ranks are not always statistically distinguishable from other beliefs. Centrality also varied across measures, indicating that different types of beliefs serve different functions within the network: egalitarianism had the highest betweenness and closeness centrality, indicating it served as a bridge connecting other beliefs, while EPA regulations had the highest strength centrality, meaning it had the strongest connections (i.e., highest partial correlations) in the network. The Bayesian network, however, complicates this picture: egalitarianism does not occupy an upstream, "parent" position in the directed network; instead, support for an international climate agreement, a secondary/policy belief, has the highest out-degree of any node. Individualism (a deep core belief) and perceived climate risk (a policy core belief), meanwhile, rank highly by both approaches. Structural importance in the climate change belief network cuts across the deep core, policy core, and secondary belief hierarchy that theory predicts, as individualism, perceived climate risk, and support for an international agreement are robustly central regardless of whether the network is modeled as undirected or directed, while egalitarianism's apparent centrality holds only in the undirected model. These findings suggest that conceptualizing belief systems as networks opens up new theoretical and empirical understanding of how different beliefs function. A network approach complements and does not displace a hierarchical one, but comparing the two reveals which beliefs' structural importance depends on modeling assumptions and which does not – a distinction a hierarchical approach alone cannot make.

## Belief Systems and Belief System Networks

Beliefs are propositions taken to be true, and the interconnections among them form a belief system. Early work on belief systems by Converse (1964) noted that belief systems consist of a "configuration of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence" (3). Constraint or interdependence among idea-elements (i.e., beliefs) within a belief system means that one belief influences others, and a change in one belief is associated with a change in another. Converse (1964) also noted that beliefs can vary by centrality, where a belief that is "central" is less likely to change than a belief that is less central. More recently, Brandt and Sleegers (2021) argued that a theory of belief systems must contain three elements: a) the components of the belief system must be connected; b) these connections should be causal; and c) contain an accounting of the potential impacts of exogenous factors.

Two approaches have developed to characterize how connections within belief systems are structured and how a belief's importance, or centrality, should be understood. The first holds that belief systems have a hierarchical structure, where higher-order beliefs, which are abstract beliefs and values, connect to and constrain lower-order beliefs about specific issues (e.g., Hurwitz and Peffley 1987, 1992; Peffley and Hurwitz 1985). Building on this work, policy scholars using the Advocacy Coalition Framework (ACF) have noted the importance of beliefs, structured hierarchically, for forming and maintaining advocacy coalitions, policy learning, and policy change (Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). Using ACF terminology, beliefs at the top of the hierarchy are known as deep core beliefs and can include political ideology, cultural worldviews, and environmental values, among others. The second tradition instead holds that belief systems are networks of interconnected beliefs, in which a belief's importance is a function of its structural position among its horizontal connections to other beliefs rather than its place in a hierarchy (Boutyline and Vaisey 2017; Brandt, Sibley, and Osborne 2019; Brandt and Sleegers 2021). Beliefs can be categorized by type (i.e., deep core, policy core, and secondary) across the two approaches, but the function of those beliefs and how they constrain other beliefs differ, with a hierarchical approach positing top-down influences and a network approach allowing influence to be determined empirically by the structure of the network.

As noted, belief types consist of deep core, policy core, and secondary beliefs. Deep core beliefs are foundational values that are sufficiently abstract that they can inform views on a variety of policy issues – beliefs about the proper role of government versus the market, the balance of expertise and centralized authority versus communal decision making, and the relative priority of values like liberty, equality, and security (Jenkins-Smith et al. 2014, 492).

Additionally, deep core beliefs are central beliefs in the Converse (1964) sense and are, therefore, resistant to change, and they act as filters for how information about policy issues is processed (Kahan 2012; Nowlin 2021). A network approach treats this claim as a testable, empirical question rather than an assumption built into the model. Deep core beliefs are important not because of their position atop a hierarchy but only if they occupy central or structurally important positions among a network's nodes. Several studies have found that they are. Using the American National Election Study (ANES), Boutyline and Vaisey (2017) found that self-reported political ideology occupies a central role in a network of beliefs about various policy issues (abortion, environment, guns, death penalty, immigration, inequality, welfare spending), and Brandt, Sibley, and Osborne (2019) similarly find that symbolic beliefs based on identities (political ideology and partisanship) are more central within belief system networks than operational (issues-based) beliefs (see also Fishman and

Davis 2022). Notably, these findings come from belief networks that span several policy issues at once; whether ideology and partisanship play the same organizing role within a belief network built around a single issue is a separate, open question I return to in the Discussion.

While deep core beliefs have been measured in various ways, increasingly scholars are using cultural worldviews developed from cultural theory to conceptualize and measure them (Hornung and Bandelow 2021; Jenkins-Smith et al. 2014; Nohrstedt et al. 2023; Ripberger et al. 2014; Sotirov and Winkel 2016). In brief, cultural theory posits four cultural worldviews, or cultural types, which are derived from ideas about how societal relationships should ideally be structured. The four cultural types include *hierarchs*, who value clear lines of authority and rules that guide individual behavior; *egalitarians*, who value equality and collective well-being; *individualists*, who value autonomy and individual choice; and *fatalists*, who see the world as capricious and beyond human control. An individual's cultural worldview has been shown to inform their views on a host of specific policy issues, including national security (Ripberger, Jenkins-Smith, and Herron 2011); gun control (Kahan and Braman 2003); trade-offs between rights and security (Jenkins-Smith and Herron 2009); immigration (Jackson 2015); vaccines (Song 2014); and climate change (Jones 2011; Kahan et al. 2012; Nowlin and Rabovsky 2020).

Following deep core beliefs are *policy core beliefs*, which involve beliefs about particular policy issues such as immigration, national security, or climate change and which the hierarchical view holds are influenced by deep core beliefs. Finally, *secondary beliefs* are beliefs regarding policy approaches, such as a carbon tax or Environmental Protection Agency (EPA) regulations, and are influenced by both deep core and policy core beliefs. A network approach retains this same vocabulary, as deep core, policy core, and secondary beliefs remain useful descriptive categories for types of nodes, but leaves open the possibility that belief centrality does not follow the specific deep core to policy core to secondary belief ordering. Instead, a belief's centrality is a function of its connections to other nodes regardless of its type or category.

The hierarchical structure of beliefs includes the necessary elements of a theory of belief systems outlined by Brandt and Sleegers (2021), as the belief system components are connected and causality runs from higher-order (deep core) to lower-order (secondary) beliefs. Yet, one issue with the hierarchical conceptualization is that the boundaries between different types of beliefs are fuzzy: I treat environmental orientation as a deep core belief because it could shape beliefs about several environmental issues and policy areas (e.g., climate change, energy production, endangered species protection), but other scholars could reasonably treat it as a policy core belief instead, with "the environment" itself seen as a policy issue. Each belief type has also been measured in different ways, leading to different conclusions among scholars about belief systems (Henry et al. 2022; Ripberger et al. 2014). A network approach avoids this problem by not assuming the vertical, hierarchical relationship among types of beliefs, but instead a belief's influence on other beliefs is based on measures of centrality rather than its place in a hierarchy.

One model of how belief system networks form begins with a single central belief that generates broader stances, which in turn stochastically generate further, more specific beliefs in a recursive process that yields a center-periphery structure (Boutyline and Vaisey 2017, 1378–79).

As the belief system forms a center-periphery structure, some beliefs can begin to act as bridges, connecting beliefs in the periphery to other beliefs in the network and keeping an otherwise potentially disparate set of beliefs connected. This is similar to the filtering role a hierarchical account attributes to deep core beliefs, but defined here by a belief's position among its connections rather than its place in a hierarchy. Multiple measures of centrality are typically used to determine the prominence of various beliefs in a belief system network (Brandt and Sleegers 2021; Camina et al. 2022; Epskamp, Borsboom, and Fried 2018; Fishman and Davis 2022; Keskintürk 2022; Turner-Zwinkels and Brandt 2022; Verschoor et al. 2020), including *betweenness*, *closeness*, and *degree (strength)*, each placing distinct emphasis on what makes a node more or less important in the network.

Betweenness centrality measures the extent to which a node lies on paths between other nodes, making beliefs high in betweenness important links, or gatekeepers, connecting the core of the network to its periphery. The betweenness centrality equation for node $n_i$ is

$$C_B(n_i) = \sum_{j<k} \frac{g_{jk}(n_i)}{g_{jk}}$$

where $g_{jk}$ is the geodesic, or shortest path between nodes $j$ and $k$, and $g_{jk}(n_i)$ is the number of shortest paths where $n_i$ is present (Luke 2015; Quintana 2023).

Closeness centrality measures how close a node is to other nodes in the network; nodes with shorter average paths to others are thought to be more influential, since they can more readily affect other beliefs through a direct effect. The equation for the closeness centrality of node $n_i$ is

$$C_C(n_i) = \left[ \sum_{j=1}^{g} d(n_i, n_j) \right]^{-1}$$

where $d$ is the distance between nodes, so closeness is the inverse of the sum of distances between $n_i$ and the other nodes in the network (Luke 2015; Quintana 2023).

Degree centrality measures the number of direct connections to a node, with more connections indicating greater prominence. Strength centrality is its weighted analogue, used when connections have a value rather than being binary: the sum of edge weights rather than the count of edges, so nodes with high strength centrality are strongly, not just widely, connected to others.

In the following analysis, I examine the network of beliefs about climate change of a quota-based sample of the US public. Next, I briefly describe several drivers of the public's views on climate change.

## Climate Change Beliefs

Even as climate scientists agree that climate change is happening, is anthropogenic, and poses significant risks, opinions are divided on each of those questions among some elites and some of the public. Several factors have been shown to influence beliefs about climate change among the public, including deep core beliefs such as partisanship, ideology, cultural worldviews, environmental attitudes, as well as policy core beliefs, including views about the scientific consensus, and secondary beliefs associated with aversions to the solutions presented to address climate change.

Perhaps the clearest divides regarding climate change are connected to political beliefs. Several studies have shown that, among the mass public in the US, conservatives and Republicans are less likely than liberals and Democrats to believe that climate change is happening or poses a risk (Bugden 2022; McCright and Dunlap 2011). Evidence suggests that polarization among the public is driven by elite polarization, as the public takes cues about policy and political issues from the political figures and news sources that they trust (Brulle, Carmichael, and Jenkins 2012; Carmichael and Brulle 2017; Krosnick et al. 2006; Merkley and Stecula 2018, 2021; Tesler 2018).

Apart from political beliefs, cultural worldviews derived from cultural theory have also been associated with climate beliefs. Early theorizing of cultural theory noted that the various cultural worldviews hold different "myths of nature" that shape how they view the natural environment and the risks posed to it: egalitarians tend to see nature as fragile, individualists as robust, hierarchs as tolerant within carefully managed limits, and fatalists as random and unmanageable (Grendstad and Selle 2000; Thompson, Ellis, and Wildavsky 1990). These myths of nature likely drive climate beliefs. Research has found that egalitarians are the most concerned about climate change and individualists the least, with hierarchs occupying a "middle" space (Jones 2011; Kahan et al. 2012; Kahan, Jenkins-Smith, and Braman 2011; Nowlin and Rabovsky 2020; Ripberger et al. 2014; Verweij et al. 2006), while fatalists, given their view of nature as capricious, tend not to have clearly defined climate beliefs (Jones 2011; Nowlin and Rabovsky 2020).

Broader environmental values and attitudes also shape views on climate change. One of the most commonly used measures of environmental orientation is the New Ecological Paradigm (NEP) scale (Dunlap 2008; Sparks et al. 2021; Stern, Dietz, and Guagnano 1995). In brief, the NEP scale measures "beliefs about humanity's ability to upset the balance of nature, the existence of limits to growth for human societies, and humanity's right to rule over the rest of nature" (Dunlap et al. 2000, 427). Using the NEP scale, a pro-environmental orientation is associated with concerns about the robustness of nature and the impacts of human activities on the natural world. Previous research has found that a pro-environmental orientation, as measured by the NEP scale, is associated with views about climate change (Nowlin 2019; Ziegler 2017).

As noted, opinions differ on climate change despite the clear scientific consensus. A series of studies have found that acceptance of the scientific consensus around climate change acts as a "gateway belief" that connects to beliefs that climate change is happening and also connects to support for actions to address climate change (Goldberg et al. 2019; S. L. van der Linden et al. 2015; S. van der Linden, Leiserowitz, and Maibach 2019). Using experimental research designs, scholars have found that those exposed to a message about the scientific consensus were more likely than those not exposed to a consensus message to change their beliefs regarding climate change (i.e., it's happening, it's anthropogenic, and it's a risk) as well as increase their support for public action (Van Der Linden 2021).

Political beliefs are a leading source of division over climate change, and one explanation for their influence is solution aversion (Campbell and Kay 2014): because Republicans/conservatives are generally more skeptical than Democrats/liberals of government's ability to address societal problems, and climate discourse centers on government action, framing climate change as requiring government solutions is likely to trigger skepticism about the solution that broadens into skepticism about the underlying problem. In essence, the solution aversion hypothesis posits that secondary beliefs and/or policy preferences drive broader views of policy issues – evidenced by the finding that the debate over the Kyoto Protocol polarized views about climate change itself (Krosnick, Holbrook, and Visser 2000).

Recent research using a network approach to examine climate change beliefs found that how worried respondents are about climate change, a policy core belief, was the most central belief in the climate change belief network (Lee et al. 2024), along with support for regulating greenhouse gas emissions (secondary belief/policy preference) and beliefs about potential harm from climate change (policy core belief). Other studies found differences in the structure of climate change belief networks across the global north and south (Lee et al. 2025), racial groups (Stewart, Rivera-Kientz, and Dacey 2024), partisan groups (Lee et al. 2024), and ideological groups in Denmark (Lind, Morton, and Dalege 2024). While these studies provide important insights into the climate change belief system of various publics, they did not include measures of many deep core beliefs shown to be important influences of climate change beliefs, including cultural worldviews and environmental orientation.

### Theoretical Expectations

Previous research points to several distinct expectations about which types of beliefs should be most structurally central in a climate change belief network, though, to my knowledge, these expectations have not been examined with measures of cultural worldviews, environmental orientation, and climate policy preferences within a single network. A hierarchical view of belief systems holds that deep core beliefs – cultural worldviews, political ideology, and pro-environmental orientation – shape and constrain other beliefs, and so should likely occupy the most central positions in the network. Work on the gateway belief model instead points to perceived scientific consensus, which is theorized to serve as a causal entry point that shifts downstream climate beliefs. A third possibility, grounded in solution aversion, is that beliefs about specific policy approaches are most central, because resistance to a policy solution can influence how the underlying problem itself is perceived. The following analysis examines centrality across all three types of beliefs simultaneously, asking which beliefs, and which types of beliefs, occupy the most structurally important positions in the network, and whether that varies depending on how centrality is measured. For example, betweenness and closeness centrality, which capture a belief's role in bridging and quickly reaching other beliefs in the network, most directly reflect the kind of constraining, foundational role a hierarchical account attributes to deep core beliefs. Strength centrality, which reflects the number and magnitude of a belief's direct connections, more directly reflects the pattern a solution-aversion account would predict for policy preferences as both deep core and policy core are likely to be connected to preferences. In the next section, I discuss the data and measurements used to explore the climate change belief network.

## Data and Analysis

To examine the belief system network around climate change, I use original public survey data collected in October 2017 from a quota-based sample (age, gender, race/ethnicity, drawn from the US Census) of about 1200 US residents, obtained through Survey Sampling International (SSI, now Dynata) and administered online through Qualtrics. Respondents were asked about their views on climate change, support for climate policies, agreement with New Ecological Paradigm (NEP) statements, cultural worldview based on cultural theory, and their ideological and partisan attachments.[1]

Although these data were collected in October 2017, several considerations support their continued relevance. Deep core beliefs such as cultural worldviews are theorized to be especially resistant to change (Converse 1964), so the overall organization of the belief network should likely be comparatively stable even as specific policy debates evolve. This is consistent with Lee et al. (2024), who find that the structure of the climate change belief network is largely consistent over time, particularly among Republicans and Democrats. It remains an open question, however, whether more recent shifts in polarization and climate discourse have altered the network's structure, and future research should replicate this analysis with more recent data to test whether the patterns reported here are enduring.

## Belief Measures

Cultural theory, political ideology, partisanship, and the NEP scale were used to measure respondents' deep core beliefs; beliefs about whether climate change is happening, its risk, and the scientific consensus measured policy core beliefs; and support for several climate policy approaches measured secondary beliefs. I describe each of the measures below.[2]

For cultural worldview measures, I rely on a standard set of survey questions used in many previous studies (e.g., Jones 2011; Moyer and Song 2019; Nowlin and Rabovsky 2020; Tumlison and Song 2019), which have been demonstrated to have convergent, discriminant, and predictive validity (Swedlow et al. 2020). Respondents rated their agreement with 12 statements (three per cultural type) on a 1 (strongly disagree) to 7 (strongly agree) scale, presented in random order; full question wording is provided in the Appendix.

For analysis, I combined and averaged the three statements into a 1 to 7 scale for each worldview, with higher scores indicating greater agreement with the statements associated with that cultural type. Each scale showed sufficient reliability: hierarchy ($M$ = 4.8, $sd$ = 1.4, Cronbach's $\alpha$ = 0.74), egalitarianism ($M$ = 4.68, $sd$ = 1.59, $\alpha$ = 0.82), individualism ($M$ = 4.48, $sd$ = 1.44, $\alpha$ = 0.76), and fatalism ($M$ = 4.04, $sd$ = 1.6, $\alpha$ = 0.8).

To measure political ideology, I asked respondents to self-identify on a 1 to 7 scale, with 1 = strongly liberal, 4 = middle of the road, and 7 = strongly conservative. The ideology scale has a $M$ = 3.99, and a $sd$ = 1.8. For partisanship, I asked respondents which party they most identified with, followed by how strongly they identified with that party. I combined the two questions into a 1 to 7 scale with 1 = strong Democrat, 4 = true independent, and 7 = strong Republican. Partisanship has a $M$ = 3.88, and a $sd$ = 1.96. Full question wording for both measures is provided in the Appendix.

[1] This study was not preregistered, and no a priori power analysis was conducted. The bootstrap correlation-stability coefficients reported in the Results section, all at or above the 0.50 threshold Epskamp et al. (2018) recommend, serve as a post hoc check that the sample size supports stable centrality estimates. The data and analysis code are organized in a version-controlled GitHub repository (hidden until publication), including the survey data, all analysis scripts, and the manuscript source, with package versions pinned via `renv` in R to support exact reproducibility.

[2] Summary statistics are shown in the appendix, Table A1.

To examine deep core beliefs related to the environment, I used three "pro-environment" statements from the NEP scale, rated on a 1 to 7 agreement scale as with the cultural worldview measures and averaged into a single 1 to 7 scale ($M$ = 5.32, $sd$ = 1.33, Cronbach's $\alpha$ = 0.75). The three statements, randomized on the survey, are listed in the Appendix.

For the policy core belief measures related to climate change, I examined whether respondents believe that climate change is happening (yes or no), and how sure they are that it is/is not happening (0 = Not at all sure, 1 = Somewhat sure, 2 = Very sure, 3 = Extremely sure). The questions, stated below, were combined into a nine-point scale with -4 indicating that respondents are *extremely sure* that climate change is *not* happening to 4 indicating that respondents are *extremely sure* that climate change *is* happening. The *happening* measure has a $M$= 2.37, and a $sd$ = 2.1.

In addition, respondents were asked about their perception of risks associated with climate change on a 0 (no risk) to 10 (extreme risk) scale, and the *risk* measure has a $M$ = 6.95 and a $sd$ = 2.76. Finally, I asked respondents their perceptions of the scientific consensus surrounding climate change. The *sciconsensus* measure is dichotomous and has a $M$ = 0.63 ($sd$ = 0.48) indicating that 63 percent of respondents believe there is a scientific consensus on climate change.[3]

It is worth noting that this operationalization of perceived scientific consensus differs from the framing used in much of the gateway belief model literature, which typically presents respondents with an explicit numerical consensus estimate (e.g., that 97 percent of climate scientists agree that climate change is happening) before asking about downstream beliefs (S. van der Linden, Leiserowitz, and Maibach 2019). The sciconsensus measure used here instead asks respondents for their own belief about whether such a consensus exists, without providing a specific consensus figure. This is a more conservative measure in the sense that it does not prime respondents with the consensus estimate itself, so it may better capture organically held beliefs about scientific consensus rather than beliefs induced by exposure to a consensus statistic. Additionally, the measure is similar to that used by Lee et al. (2024).

Finally, for secondary beliefs/policy preferences, I measured support for seven policy approaches (an international agreement, renewable energy, nuclear energy, a carbon tax, EPA regulations, cap-and-trade, and geoengineering) using a 1, strongly oppose, to 7, strongly support scale, presented in random order; full question wording is provided in the Appendix. Support was highest for renewable energy and lowest for nuclear energy, with EPA regulations, geoengineering, an international agreement, a carbon tax, and cap-and-trade falling in between in that order; full descriptive statistics for all variables are reported in Table A1.

## Belief System Network Analysis

In a belief system network, nodes represent observed variables (i.e., beliefs), and edges represent the statistical relationships between them, weighted by the strength of that relationship (Epskamp, Borsboom, and Fried 2018). Belief system networks are typically exploratory and therefore often assumed to be undirected (Costantini et al. 2015). However, in the following analysis I examine both the undirected and directed networks.

[3]The `sciconsensus` dummy variable is coded 1 = "Most scientists think climate change is happening" and 0 = all other responses (disagreement among scientists, most scientists think it is not happening, or don't know). Full question wording and variable labels for all three measures are provided in the Appendix.

First, I estimate the climate change belief system network using a Gaussian graphical model (GGM), a type of pairwise Markov random field (PMRF) model commonly used to examine psychological networks (Epskamp 2017). In a GGM, the edges represent partial correlation coefficients estimating the association between nodes while controlling for the rest of the network, based on the inverse of the covariance matrix (Epskamp and Fried 2018); polychoric correlation is used for the ordinal variables in the data. The network is then regularized using the "least absolute shrinkage and selection operator" (LASSO), which reduces the possibility of spurious correlations by shrinking small coefficients to essentially zero (Brandt, Sibley, and Osborne 2019; J. Friedman, Hastie, and Tibshirani 2008), with the LASSO's tuning parameter selected via the extended Bayesian Information Criterion (EBIC), which penalizes model complexity (Epskamp 2017; Foygel and Drton 2010). I estimate and illustrate the network using the `bootnet` and `qgraph` packages in `R` (Epskamp et al. 2012; Epskamp and Fried 2023), calculating betweenness, closeness, and strength centrality through a bootstrapping procedure using 1000 resamples, which yields the estimated centrality and 95% confidence interval for each measure (see Epskamp and Fried 2023, 2018).

The GGM is an undirected model: it estimates whether two beliefs are conditionally related but not the direction of that relationship, the standard approach in exploratory belief-network research where strong a priori commitments about causal precedence are typically unavailable (Epskamp, Borsboom, and Fried 2018). Because a central question in this paper is whether beliefs are better understood as organized hierarchically or as a network, I therefore also estimate a companion Bayesian network (BN) using the same 17 belief nodes, to assess whether a belief's estimated centrality depends on this choice of an undirected model.[4] Unlike the GGM, a BN is a directed acyclic graph in which edges represent an estimated ordering between beliefs (Koller and Friedman 2010; Pearl 2014), offering a way to assess whether a belief occupies an upstream position relative to others – the pattern a hierarchical account would predict for deep core beliefs in particular. I learn the network structure using hill-climbing, tabu search, and constraint-based (PC) algorithms, each optimized via BIC or conditional independence testing (Scutari, Graafland, and Gutiérrez 2019), and assess stability with a nonparametric bootstrap (500 resamples), retaining edges and their estimated direction that appear in at least 50% of resamples to construct an averaged network (N. Friedman, Goldszmidt, and Wyner 2013); robustness to missing data is checked via multiple imputation ($m = 5$) (Buuren and Groothuis-Oudshoorn 2011). For each node, I calculate its out-degree (the number of other beliefs relative to which it occupies an upstream position) and Markov blanket size (the parents, children, and co-parents needed to fully predict it) (Pearl 2014) as BN analogues to the GGM's centrality measures.

These algorithms are applied to cross-sectional, purely observational survey data, with no temporal ordering, experimental manipulation, or causal-identification assumption built into the analysis, so the estimated edge directions are not causally identified.[5] "Upstream" and "downstream" positions below should therefore be read as a property of how a belief's distribution relates to others' under the model's search criteria, not as evidence that one belief *causes* another.

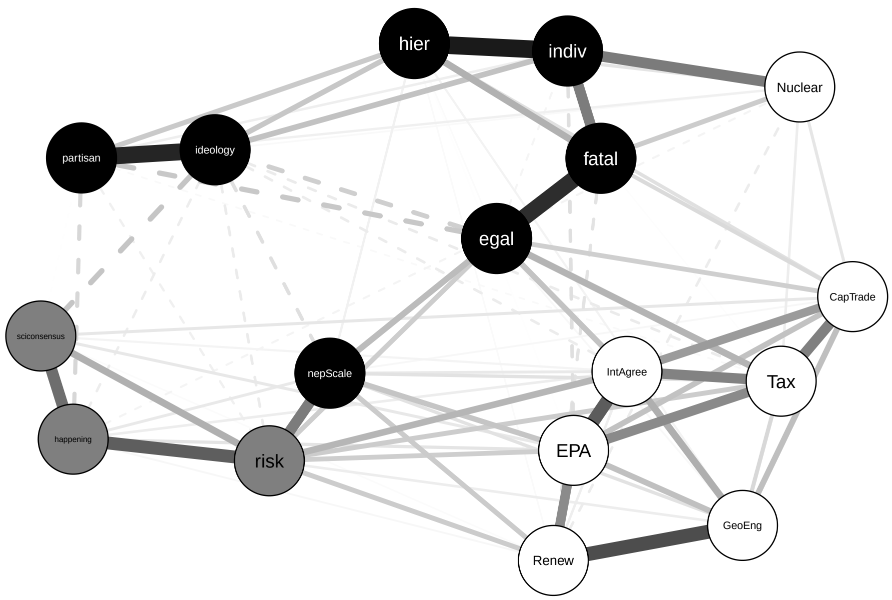
[4] Bayesian network structure learning was conducted using the `bnlearn` package in `R` (Briganti, Scutari, and McNally 2023; Scutari 2010). Continuous variables were discretized into tertiles (or, for `happening` and `risk`, substantively meaningful categories) for the score-based algorithms. Additionally, a Gaussian approximation avoiding discretization was run as a sensitivity check.

[5] Score-based algorithms such as hill-climbing and tabu search return a DAG even when several directed graphs fit the data equally well (i.e., are Markov equivalent) (Chickering 2002); the direction chosen is whichever fits the search criterion marginally better, not one the data have distinguished from its reverse. Constraint-based algorithms such as PC can identify some edge directions unambiguously but generally leave others unresolved absent further assumptions (Spirtes, Glymour, and Scheines 2000).

# Results

The results for the GGM climate change belief network include deep core beliefs (cultural worldviews, ideology, partisanship, NEP scale), policy core beliefs (climate change is happening, climate change risk, scientific consensus), and secondary beliefs (support for various policy measures), and are shown in Figure 1. Deep core beliefs are in black, policy core beliefs are in gray, and secondary beliefs are in white. Additionally, positive partial correlations between nodes are represented by a solid line, and negative partial correlations are represented by a dashed line. Finally, the network is weighted, and the strength of the partial correlations is illustrated by the thickness of the edges, with thicker paths illustrating stronger correlations.

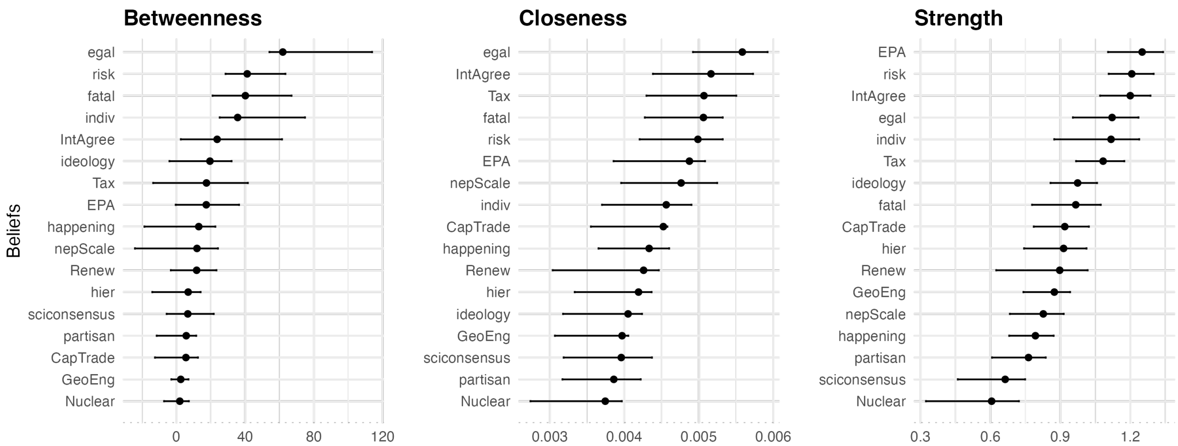
Figure 1: Belief System Network




Overall, the network has 17 nodes and 85 (of 136) non-zero edges. As shown in Figure 1, the cultural worldview, political belief, and policy preference measures each cluster together (with the exception of nuclear energy), and the NEP scale clusters with the policy core measures; deep core beliefs are associated with climate beliefs, and both are associated with support for climate policies, as expected. The strongest associations are between partisanship and ideology, between the belief that climate change is happening and both scientific-consensus beliefs and risk perceptions, among the policy measures (particularly renewable energy and geoengineering), and among the cultural worldviews (hierarchy with individualism, egalitarianism with fatalism). After visualizing the network, I next examine various centrality measures within it; results from the bootstrap procedure, including 95% confidence intervals, are shown in Figure 2.

Looking first at betweenness centrality, as shown in Figure 2, egalitarianism has the highest point estimate, with 61.89, followed by climate change risk perceptions (41.19), fatalism (40.11), individualism (35.62), and support for an international agreement (23.67). Nodes high in betweenness sit on the shortest path between other nodes in the network and act as a bridge between beliefs, so egalitarianism works to connect beliefs in

Figure 2: Measures of Network Centrality




the network. To assess whether these point estimates reflect statistically reliable rank differences, I conducted pairwise bootstrap difference tests for all 136 possible node pairs on each centrality measure (see Epskamp and Fried 2023). For betweenness, only 31 of the 136 comparisons were significant overall, and none of the 10 comparisons among these five top-ranked nodes reached significance. Egalitarianism's rank as most central on betweenness is therefore a point estimate rather than a statistically significant difference from climate change risk, fatalism, individualism, or support for an international agreement. A case-dropping bootstrap further shows betweenness has a correlation stability (CS) coefficient of 0.52, meeting the 0.50 threshold Epskamp et al. (2018) consider stable.

Egalitarianism also has the highest closeness centrality, with 0.00558, followed by support for an international agreement (0.00516), a carbon tax (0.00507), fatalism (0.00506), and climate change risk (0.00499). Beliefs high in closeness are more closely connected to other nodes, giving them shorter paths to the rest of the network and therefore greater potential influence. As with betweenness, most of these ranks are not statistically distinguishable: only 1 of the 10 comparisons among the top five closeness nodes reached significance. Egalitarianism's closeness centrality was significantly higher only than fatalism's, and not significantly different from that of support for an international agreement, a carbon tax, or climate change risk. Closeness did show a higher overall rate of significant differences than the other two measures (54 of 136 pairs), suggesting closeness centrality is more reliably differentiated across beliefs even though the top-ranked cluster itself remains largely indistinguishable. Closeness also showed a stable CS coefficient of 0.75.

Turning to strength centrality, which reflects the sum of the absolute values of a node's weighted connections, support for EPA regulations has the highest point estimate, with 1.25, followed by perceptions of climate change risk (1.21) and support for an international agreement (1.2). The EPA node's strongest partial correlations were with the other policy support variables – international agreement ($r$ = 0.28), carbon tax ($r$ = 0.201), and renewable energy ($r$ = 0.201) – as well as the NEP scale ($r$ = 0.099). Support for an international agreement was likewise most strongly correlated with the other policy support measures, including EPA regulations (0.28), a carbon tax (0.216), cap-and-trade (0.172), and geoengineering (0.138), as well as climate change risk (0.127) and egalitarianism (0.104). Climate change risk, in turn, was most highly correlated with the other policy core beliefs, including whether climate change is happening and whether there is a scientific consensus on it. These rankings should also be read cautiously as 82 of 136 pairwise comparisons were significant for strength overall, but only 1 of 10 comparisons among the top five nodes reached significance. The strength ranking of EPA regulations, climate change risk, and support for an international agreement should be read with the same caution. Strength's CS coefficient is 0.75, also exceeding the 0.50 stability threshold. The reported network is robust more broadly: re-estimating across a range of EBIC tuning values (gamma = 0 to 1) shows non-zero edges ranging from 79 to 105, but the node with the highest point-estimated strength, betweenness, and closeness centrality is unchanged at every value tested, and a cross-validation-selected network (the `huge` R package) has a materially similar level of sparsity (87 vs. 85 EBIC-selected edges).

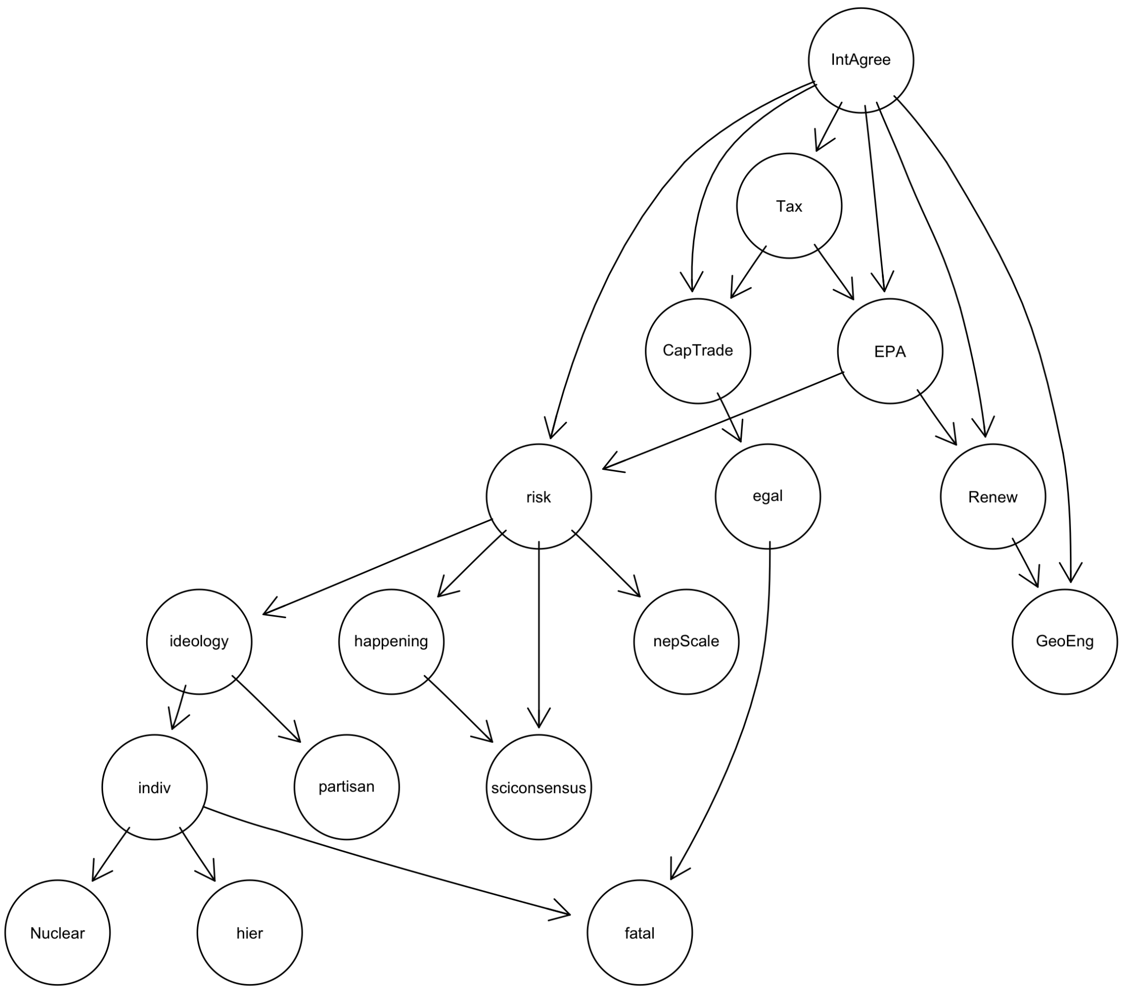
## Bayesian Network

To assess whether these conclusions depend on the choice of an undirected GGM, I estimated a companion Bayesian network using the same 17 belief nodes. Figure 3 shows the averaged, bootstrap-stable directed network (threshold = 0.50, 23 arcs retained, drawn from 27 hill-climbing arcs, 28 tabu-search arcs, and 26 PC-algorithm arcs). 8 arcs were recovered by all three algorithms, and 20 arcs were stable across all five multiply-imputed datasets.

As shown in Figure 3, support for an international agreement is the node with no in-degree connections and the largest number of out-degree connections. Additionally, Figure 3 shows that the network seems to move downward in layers rather than outward from a single node. This is an ordering estimated from arc direction, not a claim about the deep core/policy core/secondary hierarchy described above; however, this estimated layering places several secondary beliefs above deep core beliefs, inverting rather than confirming that theoretical hierarchy. Of note, other secondary beliefs, including support for a carbon tax, cap-and-trade, EPA regulations, renewable energy, and geoengineering sit just below support for an international agreement. Several of the secondary beliefs are in turn upstream of climate change risk, which sends arcs to four other nodes, including political ideology and the belief that climate change is happening. Egalitarianism, by contrast, occupies a comparatively peripheral position in the Bayesian network, as it receives a single arc, from EPA regulations, and sends a single arc, to fatalism. Perceived scientific consensus sits at the bottom of the graph as well, receiving arcs from climate change risk and the belief that climate change is happening but sending none of its own, a result similar to its low centrality in the undirected GGM.

Table 1 reports, for each node, its position in the directed network (in-degree, out-degree, and Markov blanket size) alongside its rank on each GGM centrality measure (1 = most central). The comparison reveals both convergence and divergence between the two approaches. Support for an international agreement (`IntAgree`), a secondary belief, has the highest out-degree in the network. This means that it occupies more of an upstream position in the network relative to other beliefs, including several deep core and policy core beliefs. However, this describes its estimated position in the graph, not a claim that it causes those other beliefs. Individualism (`indiv`) and perceived climate risk (`risk`) also rank highly on both GGM centrality and BN out-degree/Markov blanket size, indicating that their structural importance is likely not an artifact of either model. Egalitarianism, by contrast, had the highest betweenness and closeness centrality in the GGM, yet it ranks well outside the top nodes on every BN measure. Ideology and partisanship rank low on nearly every

Figure 3: Averaged Bayesian Network (bootstrap threshold = 0.50)



measure in both models as well: ideology's highest GGM rank is 7th (on betweenness and strength), and partisanship never rises above 13th on any GGM measure, while partisanship also has the smallest Markov blanket and no outgoing arcs in the BN.

Table 1: GGM Centrality Ranks vs. Bayesian Network Structural Position

| Node | Type | BN In-Degree | BN Out-Degree | BN Markov Blanket | GGM Betweenness Rank | GGM Closeness Rank | GGM Strength Rank |
|---|---|---|---|---|---|---|---|
| IntAgree | Secondary | 0 | 6 | 6 | 5 | 2 | 3 |
| risk | Policy Core | 2 | 4 | 6 | 3 | 5 | 2 |
| indiv | Deep Core | 1 | 3 | 5 | 2 | 8 | 6 |
| EPA | Secondary | 2 | 2 | 4 | 6 | 7 | 1 |
| ideology | Deep Core | 1 | 2 | 3 | 7 | 14 | 7 |
| Tax | Secondary | 1 | 2 | 3 | 7 | 3 | 5 |
| Renew | Secondary | 2 | 1 | 3 | 9 | 13 | 12 |
| CapTrade | Secondary | 2 | 1 | 3 | 13 | 10 | 9 |
| egal | Deep Core | 1 | 1 | 3 | 1 | 1 | 4 |
| happening | Policy Core | 1 | 1 | 2 | 11 | 9 | 14 |
| fatal | Deep Core | 2 | 0 | 2 | 4 | 4 | 8 |
| sciconsensus | Policy Core | 2 | 0 | 2 | 10 | 12 | 16 |
| GeoEng | Secondary | 2 | 0 | 2 | 11 | 16 | 11 |
| hier | Deep Core | 1 | 0 | 1 | 13 | 11 | 10 |
| nepScale | Deep Core | 1 | 0 | 1 | 13 | 6 | 13 |
| partisan | Deep Core | 1 | 0 | 1 | 13 | 15 | 15 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Nuclear | Secondary | 1 | 0 | 1 | 13 | 17 | 17 |

## Discussion and Conclusion

Climate change is a polarized issue among the US public, and beliefs about climate change form an interconnected belief system. One way to understand belief systems is through a hierarchical approach in which foundational, abstract beliefs influence other beliefs about specific issues and policy approaches. Another way to understand the structure of belief systems is as a network, with beliefs as nodes and connections between beliefs as edges.

In this paper, I used original survey data from a quota-based sample of the US public to examine the belief system network associated with climate change. Additionally, I drew on the leading influences of climate change views identified in the literature, including political beliefs, cultural worldviews, pro-environment orientation, acceptance of the scientific consensus, and solution aversion, as theoretical anchors for exploring which beliefs would be the most central in the network using the common measures of centrality, including betweenness, closeness, and strength. To assess whether these patterns depend on treating the belief system as an undirected network, I also estimated a directed Bayesian network over the same beliefs and compared centrality rankings and estimated graph position across both models.

Examining climate beliefs as a network allows an understanding of which beliefs are structurally important, rather than as a way to test specific hypotheses. The results showed that individualism, support for an international agreement, and perceived climate risk each occupy central positions on at least one measure of centrality across the undirected and directed networks. This pattern is broadly consistent with theoretical accounts that emphasize deep core beliefs and policy-specific solution aversion as organizing features of belief systems. At the same time, the pairwise comparisons show that many of these point-estimate differences are not statistically distinguishable from one another. For example, egalitarianism's betweenness centrality is not significantly different from that of individualism, fatalism, perceived risk, or support for an international agreement. Structural importance in the climate change belief network is therefore likely concentrated among a cluster of beliefs spanning multiple theoretical categories, rather than residing in any single belief or type of belief. The cluster of beliefs that are central in both the GGM and the Bayesian network includes individualism (`indiv`), perceived climate risk (`risk`), and support for an international climate agreement (`IntAgree`); egalitarianism, by contrast, is prominent only in the GGM.

The results of the GGM also showed that centrality differed across measures, with egalitarianism ranking highest on betweenness and closeness, and EPA regulations ranking highest on strength, with climate risk and support for an international agreement close behind on both. Taken on its own, this is consistent with a hierarchical account for egalitarianism, a deep core belief acting as a bridge that connects and constrains other beliefs – but the strength-centrality prominence of secondary beliefs like EPA regulations and support for an international agreement does not fit a hierarchical account, which would place such beliefs on the periphery, and instead fits a solution-aversion account, in which opposition to policy solutions shapes perceptions of the underlying problem (Campbell and Kay 2014). The Bayesian network complicates the egalitarianism reading further and only partly corroborates the solution-aversion reading. In the BN, egalitarianism has neither a high out-degree nor a large Markov blanket, so it does not occupy an upstream position; its GGM betweenness centrality more likely reflects its role bridging the cluster of cultural worldview items than the directional, constraining role a hierarchical account attributes to deep core beliefs. A stronger challenge to a hierarchical reading comes from support for an international climate agreement, a secondary/policy-level belief with the highest out-degree of any node and, tied with perceived climate risk, the largest Markov blanket – a more upstream estimated position than every deep core and policy core belief, including egalitarianism. EPA regulations' own directed-network position, by contrast, is comparatively modest (out-degree = 2, Markov blanket size = 4), so its GGM strength-centrality prominence is more evident in the undirected than the directed model, and a hierarchical account offers no obvious explanation for why a specific climate policy preference would occupy the most upstream position in the directed network. At the same time, individualism and perceived climate risk are prominent by both approaches, so the evidence does not simply favor a network account over a hierarchical one; whether a given belief's position is consistent with a hierarchical account appears to depend on the specific belief and the specific model used to estimate it, not on a general property of hierarchical versus network structure. Future research should use panel data or explicit causal-identification designs to test whether these estimated positions reflect actual causal precedence, and to clarify when a directed or undirected framework is more appropriate.

As noted earlier, this analysis examined a belief network built around a single issue, climate change – a scope choice with implications for deep core beliefs specifically. Prior belief network research spanning multiple policy issues has found ideology and partisanship to be highly central (Boutyline and Vaisey 2017; Brandt, Sibley, and Osborne 2019), but as reported above, neither ranks among the most central beliefs on any measure here, likely because a single issue gives these deep core beliefs less organizing work to do than issue-specific beliefs can do instead. This pattern is not uniform across every deep core belief, however: individualism, and to a lesser extent egalitarianism, remain prominently central despite the single-issue scope, so which deep core belief is doing the organizing – a cultural worldview versus a political identity – may matter as much as how many issues a network spans. This also suggests a hierarchical account may be better suited to multi-issue belief systems than to single-issue ones like this one, though that remains an empirical question for future research.

Another theoretical anchor used in this paper was the gateway belief model, where agreement with the scientific consensus about climate change informs other climate beliefs. However, neither type of network indicated structural importance for the consensus belief. The implication of this finding deserves a careful interpretation. The gateway belief model is a causal, interventional framework. It is typically tested using experimental or longitudinal designs that trace how exposure to a consensus message shifts later beliefs (Said, Frauhammer, and Huff 2022). The present analysis estimates belief networks using cross-sectional data, and centrality in these types of networks reflects structural embeddedness, not causal influence. A belief can be a causal entry point for belief change without also being central to how beliefs are organized at a given point in time. Recent work also shows that the gateway belief model's effects are conditional rather than universal, likely varying with respondents' engagement with the consensus message (Said, Frauhammer, and Huff 2023) and with the specific mechanisms and boundary conditions of the messaging itself (Göbel et al. 2026). The low centrality of scientific consensus in the network analysis, then, does not contradict the gateway belief model. It is likely, instead, that consensus beliefs are not structurally embedded at the core of the broader climate belief system, even though they can still function as effective causal mechanisms under targeted informational interventions.

This conclusion is reinforced by the Bayesian network, where perceived scientific consensus sits at the bottom of the estimated graph, sending no arcs of its own and receiving arcs only from climate risk and the belief that climate change is happening, which is similar to the peripheral position it occupies in the undirected model. Future research should combine network and experimental approaches to better understand the structural and causal accounts of the gateway belief model.

While these results shed light on the structural importance of various beliefs within the climate change belief network, several avenues for future research remain. The results shown here illustrate the centrality of beliefs across the entire sample, but it is possible that subgroups have distinct belief networks. Prior work has found that belief-network structure varies across ideological and demographic subgroups generally (Camina et al. 2022) and, specific to climate change, across partisan and racial/ethnic subgroups (Lee et al. 2024; Stewart, Rivera-Kientz, and Dacey 2024). Additional research with a larger sample size should examine possible differences in networks across the various cultural types.

These findings speak to scholars, most notably those doing policy process research, who have largely relied on a hierarchical conception of belief systems with deep core beliefs constraining policy core and secondary beliefs. The results here suggest a network approach offers analytic leverage the hierarchical framework alone may not capture: structural importance in the climate change belief network does not track the deep core/policy core/secondary ordering a hierarchical account predicts, with individualism, perceived climate risk, and support for an international agreement robustly central regardless of whether the network is modeled as undirected or directed, and support for an international agreement occupying the most upstream position of any node in the directed network. Rather than replacing the hierarchical approach, these results suggest policy process scholars would benefit from treating belief systems as networks alongside it, particularly when the goal is to identify which beliefs are most structurally consequential rather than assuming that answer from belief type alone – with the caveat, noted above, that hierarchy may still be the more apt starting point for the multi-issue belief systems much of policy process research traditionally examines. Doing so also means pairing undirected and directed operationalizations, since the directed model missed egalitarianism's bridging role, while the undirected model has no way to express the kind of upstream, directional position support for an international agreement holds in the directed model.

For climate communication and public opinion research, the finding that perceived scientific consensus is not structurally central, together with the reframing of the gateway belief model discussed above, suggests that consensus messaging campaigns may be better understood as targeted causal interventions than as efforts to shift a structurally central belief. This distinction could have direct implications for how such campaigns are designed and evaluated.

# Appendix

The descriptive statistics for each of the variables are presented in Table A1.

Table A1: Descriptive Statistics for Belief Measures

| | Mean | SD | Min | Max |
|---|---|---|---|---|
| hier | 4.80 | 1.40 | 1.00 | 7.00 |
| egal | 4.68 | 1.59 | 1.00 | 7.00 |
| indiv | 4.48 | 1.44 | 1.00 | 7.00 |
| fatal | 4.04 | 1.60 | 1.00 | 7.00 |
| ideology | 3.99 | 1.80 | 1.00 | 7.00 |
| partisan | 3.88 | 1.96 | 1.00 | 7.00 |
| nepScale | 5.32 | 1.33 | 1.00 | 7.00 |
| happening | 2.37 | 2.10 | −4.00 | 4.00 |
| risk | 6.95 | 2.76 | 0.00 | 10.00 |
| sciconsensus | 0.63 | 0.48 | 0.00 | 1.00 |
| IntAgree | 5.07 | 1.73 | 1.00 | 7.00 |
| Renew | 5.89 | 1.39 | 1.00 | 7.00 |
| Nuclear | 4.09 | 1.92 | 1.00 | 7.00 |
| Tax | 4.79 | 1.84 | 1.00 | 7.00 |
| EPA | 5.35 | 1.66 | 1.00 | 7.00 |
| CapTrade | 4.75 | 1.71 | 1.00 | 7.00 |
| GeoEng | 5.32 | 1.55 | 1.00 | 7.00 |

## Survey Question Wording

**Cultural worldviews.** Respondents rated their agreement with the following 12 statements on a 1 (strongly disagree) to 7 (strongly agree) scale, presented in random order.

*Hierarchy:*

- *The best way to get ahead in life is to work hard to do what you are told to do*
- *Society is in trouble because people do not obey those in authority*
- *Society would be much better off if we imposed strict and swift punishment on those who break the rules*

*Egalitarianism:*

- *What society needs is a fairness revolution to make the distribution of goods more equal*
- *Society works best if power is shared equally*
- *It is our responsibility to reduce differences in income between the rich and the poor*

*Individualism:*

- *Even if some people are at a disadvantage, it is best for society to let people succeed or fail on their own*
- *Even the disadvantaged should have to make their own way in the world*
- *We are all better off when we compete as individuals*

*Fatalism:*

- *The most important things that take place in life happen by chance*
- *No matter how hard we try, the course of our lives is largely determined by forces beyond our control*
- *For the most part, succeeding in life is a matter of chance*

**Political ideology and partisanship.**

- **ideology**: *On a scale of political ideology, individuals can be arranged from strongly liberal to strongly conservative. Which of the following best describes your views?*
- **partisan**: *With which political party do you most identify? Do you completely, somewhat, or slightly identify with that party?*

**New Ecological Paradigm scale.** Randomized on the survey.

- *The balance of nature is very delicate and easily upset*
- *Humans live on a planet with very limited room and resources*
- *Humans are seriously abusing the environment*

**Climate policy core beliefs.**

- **happening**: *Do you think that climate change is happening? How sure are you that climate change [is/is not] happening?*
- **risk**: *On the scale from zero to ten, where 0 means no risk and 10 means extreme risk, how much risk do you think climate change poses for people and the environment?*
- **sciconsensus**: *Which of the following statements comes closest to your view? 1 – Most scientists think climate change is happening, 2 – There is disagreement among scientists, 3 – Most scientists think climate change is not happening, 4 – Don't know.*

**Climate policy preferences.** Presented in random order, following the prompt: *Various policy solutions have been proposed as ways for the federal government in the United States to deal with climate change. Using a one to seven scale, where 1 is strongly oppose and 7 is strongly support, please indicate your support for each of the following policy proposals.*

- **IntAgree**: *Accept internationally-established limits on U.S. production of carbon dioxide and other greenhouse gases thought to cause climate change*
- **Renew**: *Encourage research and development of renewable energy sources like solar or wind power*
- **Nuclear**: *Expand the use of nuclear energy*
- **Tax**: *A carbon tax that imposes a charge on coal, oil, and natural gas in proportion to the carbon they contain*
- **EPA**: *Environmental Protection Agency regulations that limit the amount of carbon dioxide and other greenhouse gases that power plants can emit*
- **CapTrade**: *A cap and trade program that sets an overall cap on emissions through allowances that can be bought, sold, or saved for future use*
- **GeoEng**: *Encourage development of "geoengineering" technology such as filters that remove excess carbon dioxide from the air and high-altitude orbiting reflectors to reduce solar heating*

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